

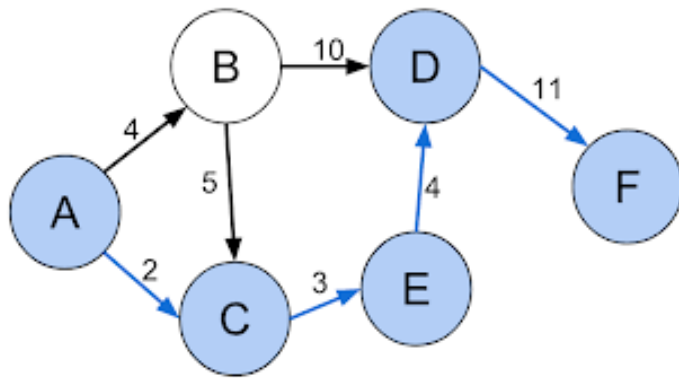
Algorithmic Game Theory

Zhengyang Liu, BIT

Administration

- Instructor: 刘正阳, zhengyang@bit.edu.cn
- Week 1-11 (15:15 Tue@F404 // Thu@F302)
- Grading: Homework (20*2) + Presentation & Report (40) + scribe (20*1)
 - By group (3~4 persons)
 - Presentation: motivation + plan
 - Report: overview + your thoughts (if any)

Examples in our book



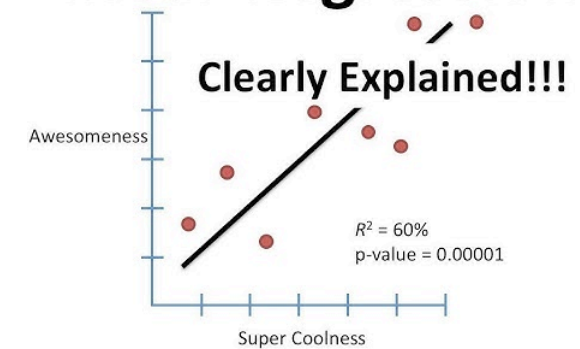
1, 3, 3, **6**, 7, 8, 9

Median = **6**

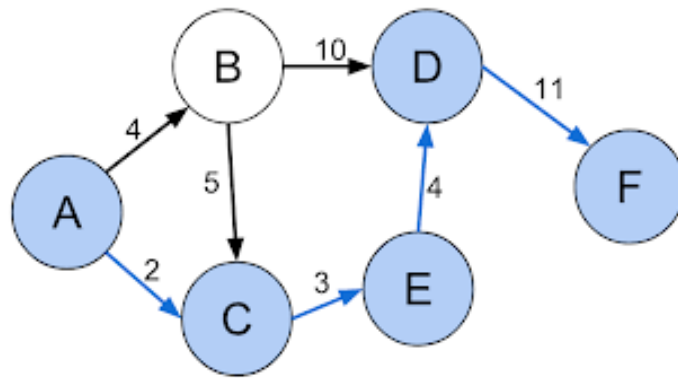
1, 2, 3, **4**, **5**, 6, 8, 9

Median = $(4 + 5) \div 2$
= **4.5**

Linear Regression



What's the difference?



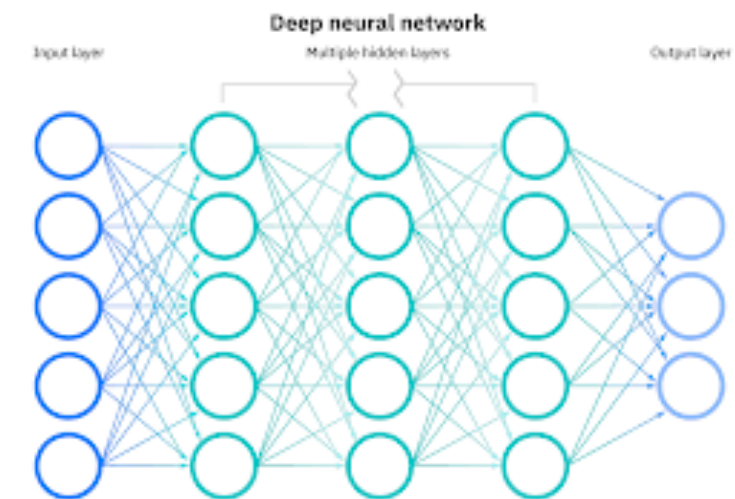
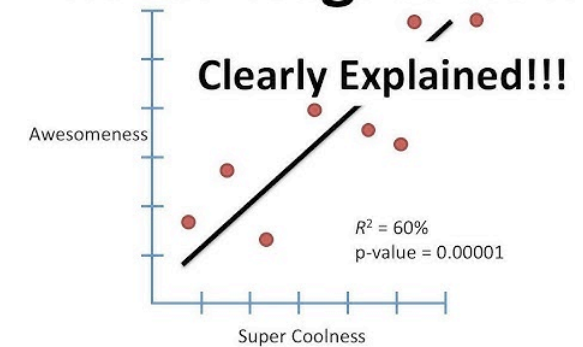
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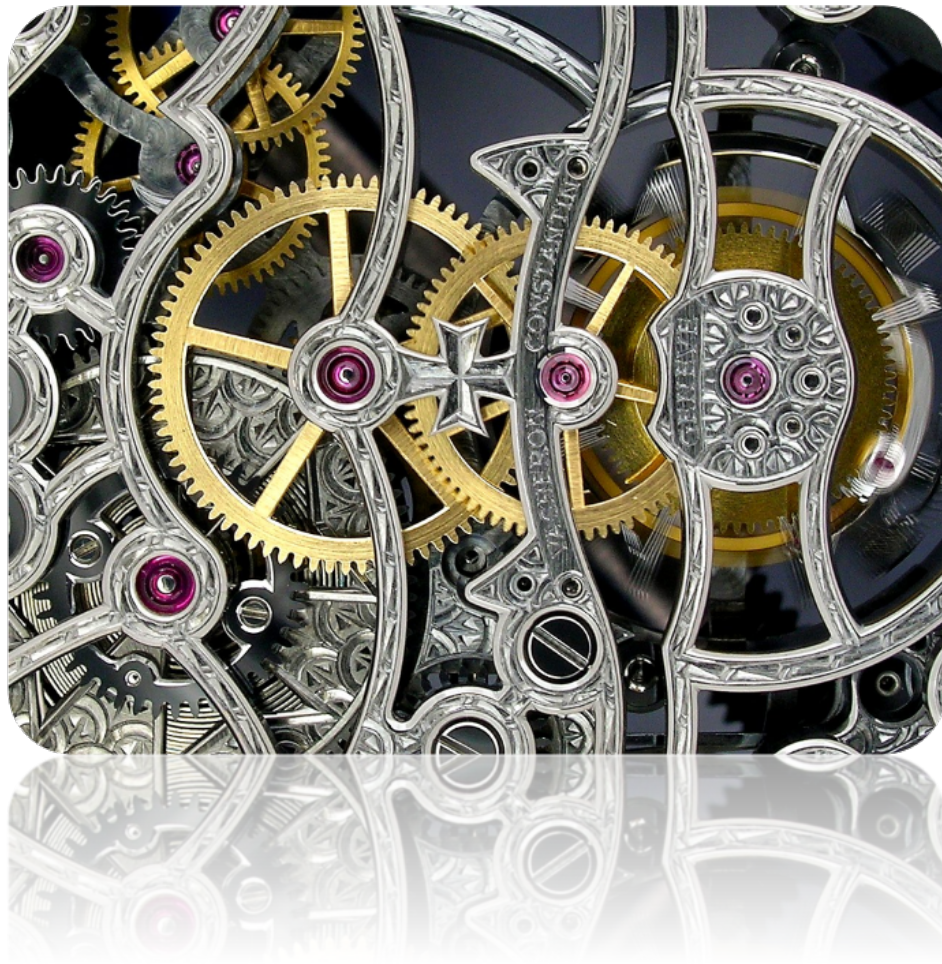
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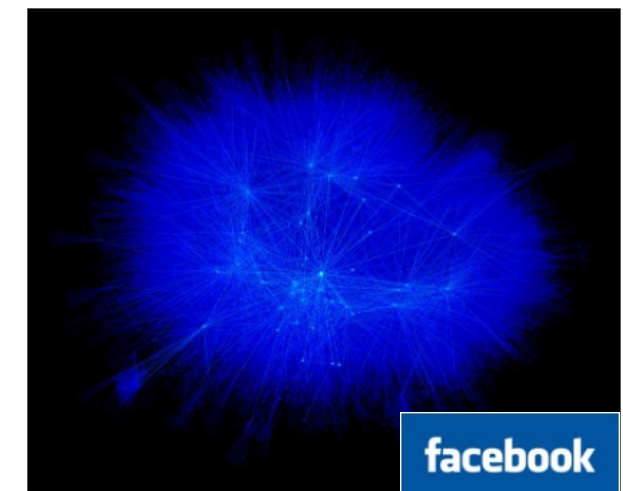
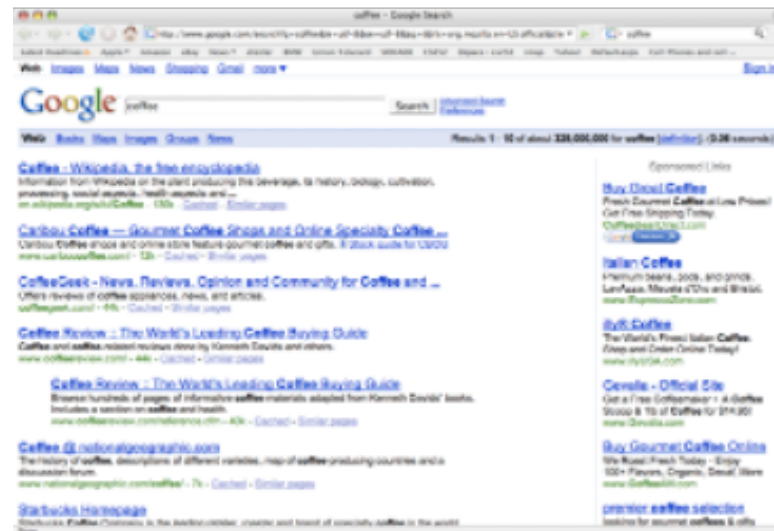
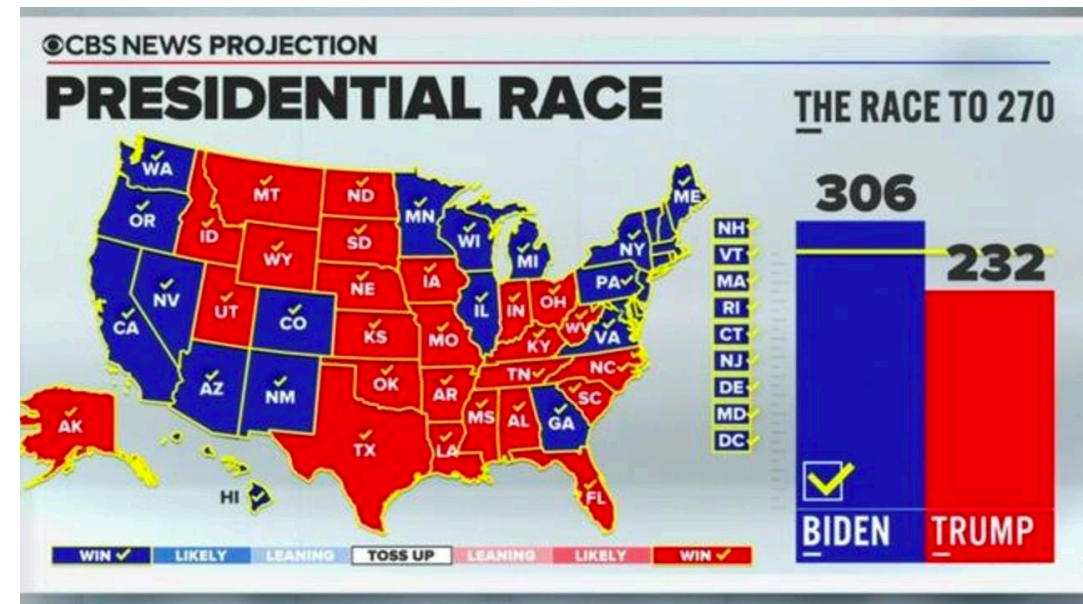


What is Algorithm?

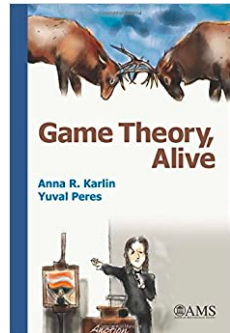
Do you know (control) everything in the system?

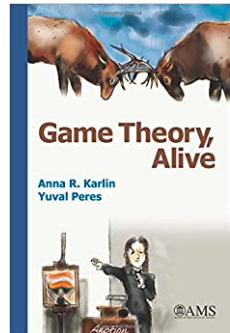


What is Game Theory?



Game Theory



- : We live in a highly connected world, with multiple self-interested agents interacting, leading to myriad opportunities for conflict and cooperation.
- **Games** are thought experiments to help us learn how to **predict rational behavior** in **situations of conflict**.
- **Situation of conflict**: Everybody's actions affect others.
- **Rational Behavior**: The players want to maximize their own expected utility.
- **Predict**: We want to know what happens in a game. Such predictions are called solution concepts (e.g., Nash equilibrium).

Guess Two-Thirds of the Average

- k players $p_1, p_2, \dots, p_k$
- Each player submits a number in $[0, 100]$, say $x_1, \dots, x_k$
- Compute $\bar{x} := \frac{1}{k} \sum_{i=1}^k x_i$
- Find x_j , closest to $\frac{2}{3}\bar{x}$

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- How about $(\frac{2}{3})^2 \cdot 100$?
- So all-zero strategy is the only NE of this game!

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- How about $(\frac{2}{3})^2 \cdot 100$?
- So all-zero strategy is the only NE of this game!
- Story: 21.6 was the winning value in a large internet-based competition organized by the Danish newspaper Politiken. This included 19,196 people and with a prize of 5000 Danish kroner.

Nash Equilibrium

PRISONER'S DILEMMA

C = cooperate D = defect (don't cooperate)

YEARS IN PRISON

	C	D
C	-1, -1	-10, 0
D	0, -10	-5, -5



	S	P	R
S	0,0	1,-1	-1,1
P	-1,1	0,0	1,-1
R	1,-1	-1,1	0,0

Informally, an **equilibrium** is a steady state. Is there a deterministic equilibrium?

We use randomness! Called **mixed strategies**.

Nash Equilibrium: A pair of strategies such that the strategy of the row player is a **Best Response** to the strategy of the column player and vice versa.

Bimatrix Games

- 2 players: **Row** and **Column**
- n, m **strategies** available
- **Payoff** matrices R, C of size $n \times m$.

$$\mathcal{G} = (R_{n \times m}, C_{n \times m})$$



payoff of the column player
for playing j
when row player plays i .

payoff of the row player for playing i
when column player plays j .

Bimatrix Games

Column player
chooses $y \in \Delta_m$

Row player
chooses $x \in \Delta_n$


$$R_{ij}, C_{ij}$$



Row gets $x^\top Ry$.
Column gets $x^\top Cy$.

Bimatrix Games

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Row gets $x^\top Ry$.
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Definition (Nash Equilibrium). (x^*, y^*) is a Nash Equilibrium iff for all possible randomized strategies x' of row player holds

$$x^{*\top} Ry^* \geq x'^\top Ry^*$$

and for all possible randomized strategies y' of column player holds

$$x^{*\top} Cy^* \geq x^{*\top} Cy'.$$

Example: Rock-Paper-Scissors

			
	0, 0	-1, 1	1, -1
	1, -1	0, 0	-1, 1
	-1, 1	1, -1	0, 0

The unique Nash Equilibrium is $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

Remark:

Contrary to Prisoner's Dilemma, in RPS **randomization** is **necessary** for Nash equilibrium to **exist**!

Example: Rock-Paper-Scissors

			
	0, 0	-1, 1	1, -1
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Nice solution concept but does it always exist?

Remark:

Contrary to Prisoner's Dilemma, in RPS *randomization* is *necessary* for Nash equilibrium to *exist*!

Example: Rock-Paper-Scissors

			
	0, 0	-1, 1	1, -1
	1, -1	0, 0	-1, 1
	-1, 1	1, -1	0, 0

von Neumann '28:

*For **two-player zero-sum games**, i.e., $R + C = 0$, it
always exists!*

Remark:

Contrary to Prisoner's Dilemma, in RPS **randomization** is **necessary** for Nash equilibrium to **exist!**

Example: Modified Rock-Paper-Scissors

			
	0, 0	-1, 1	2, -1
	1, -1	0, 0	-1, 1
	-2, 1	1, -1	0, 0

Example: Modified Rock-Paper-Scissors

			
	0, 0	-1, 1	2, -1
	1, -1	0, 0	-1, 1
	-2, 1	1, -1	0, 0

Not zero sum anymore!

Example: Modified Rock-Paper-Scissors

	25% 	50% 	25% 
33.3% 	0, 0	-1, 1	2, -1
33.3% 	1, -1	0, 0	-1, 1
33.3% 	-2, 1	1, -1	0, 0

Not zero sum anymore!

John Nash '51:

There always **exists** a **Nash equilibrium** (finite games)!

Cool but Algorithmic?



Question: Can we predict what will happen in a **large** system?

- ***Computing Nash Equilibrium:*** Design fast Algorithms to compute Nash Equilibrium!
- ***Mechanism Design:*** Design a system that will be used by users to optimize our objectives!

Cool but Algorithmic?



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Game theory: Yes, **via solution concept** (system will reach equilibrium).

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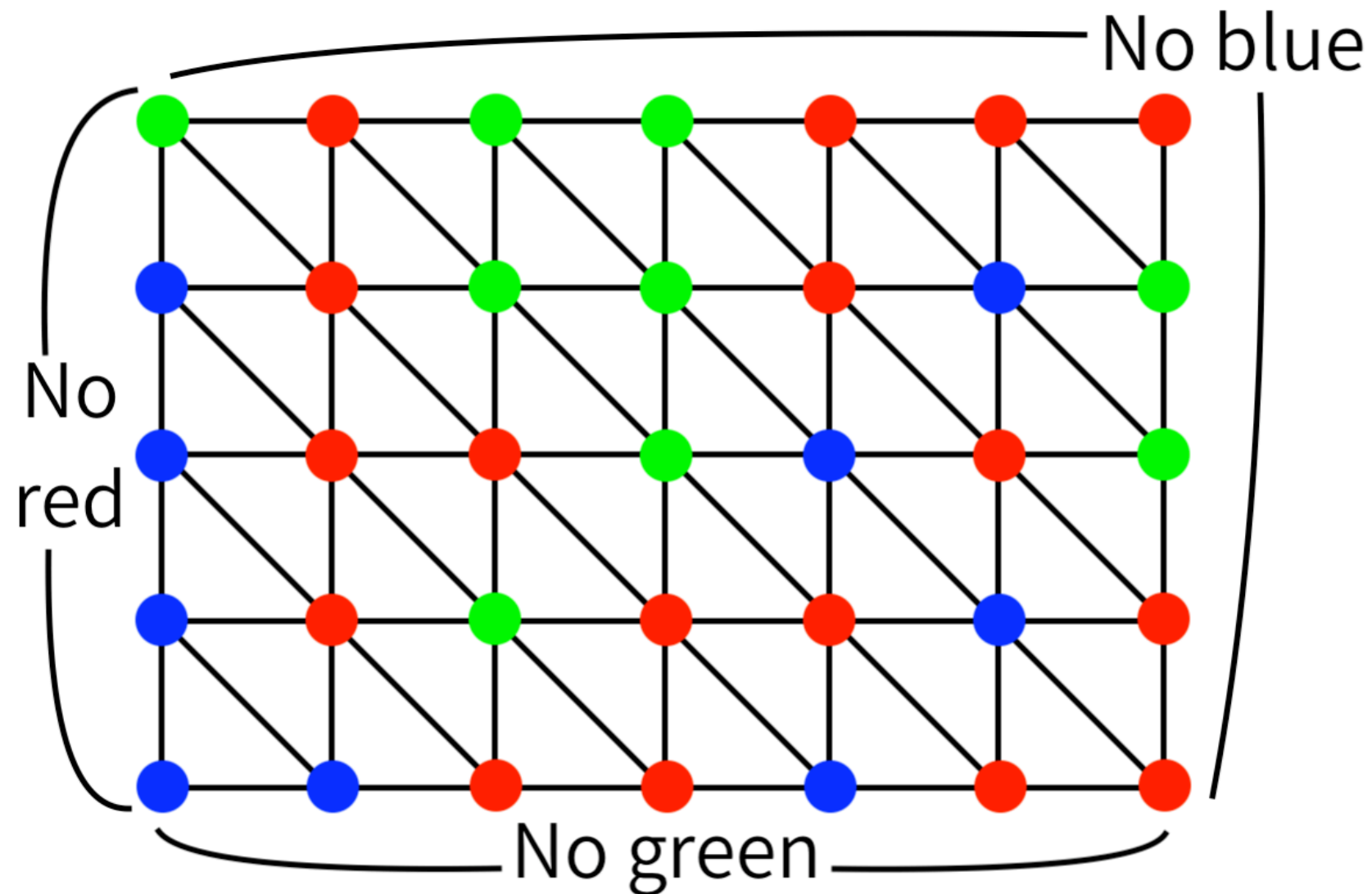
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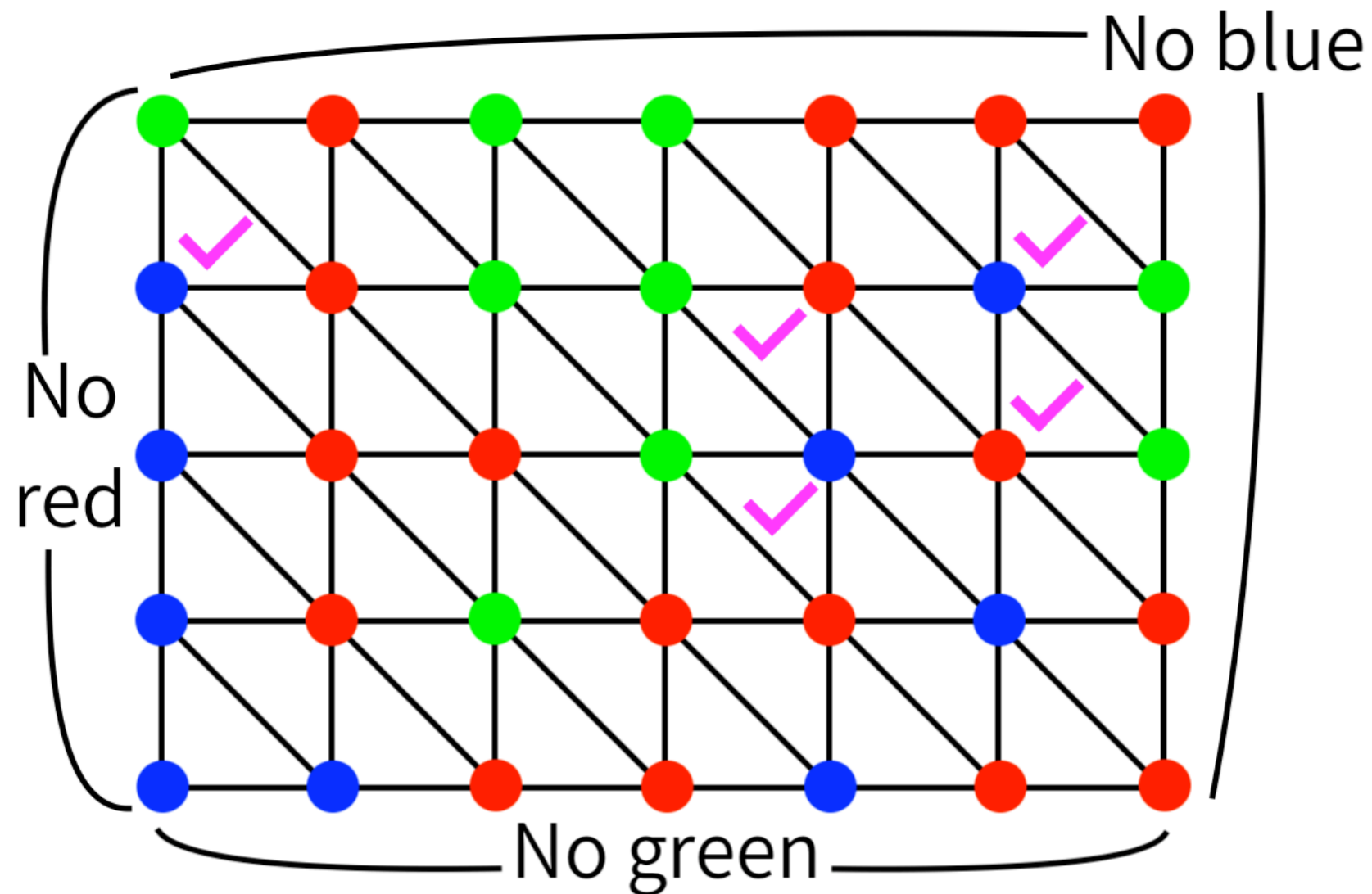
Question: How to **compute efficiently** an equilibrium?

- ***Computing Nash Equilibrium:***
Design fast Algorithms to compute Nash Equilibrium!
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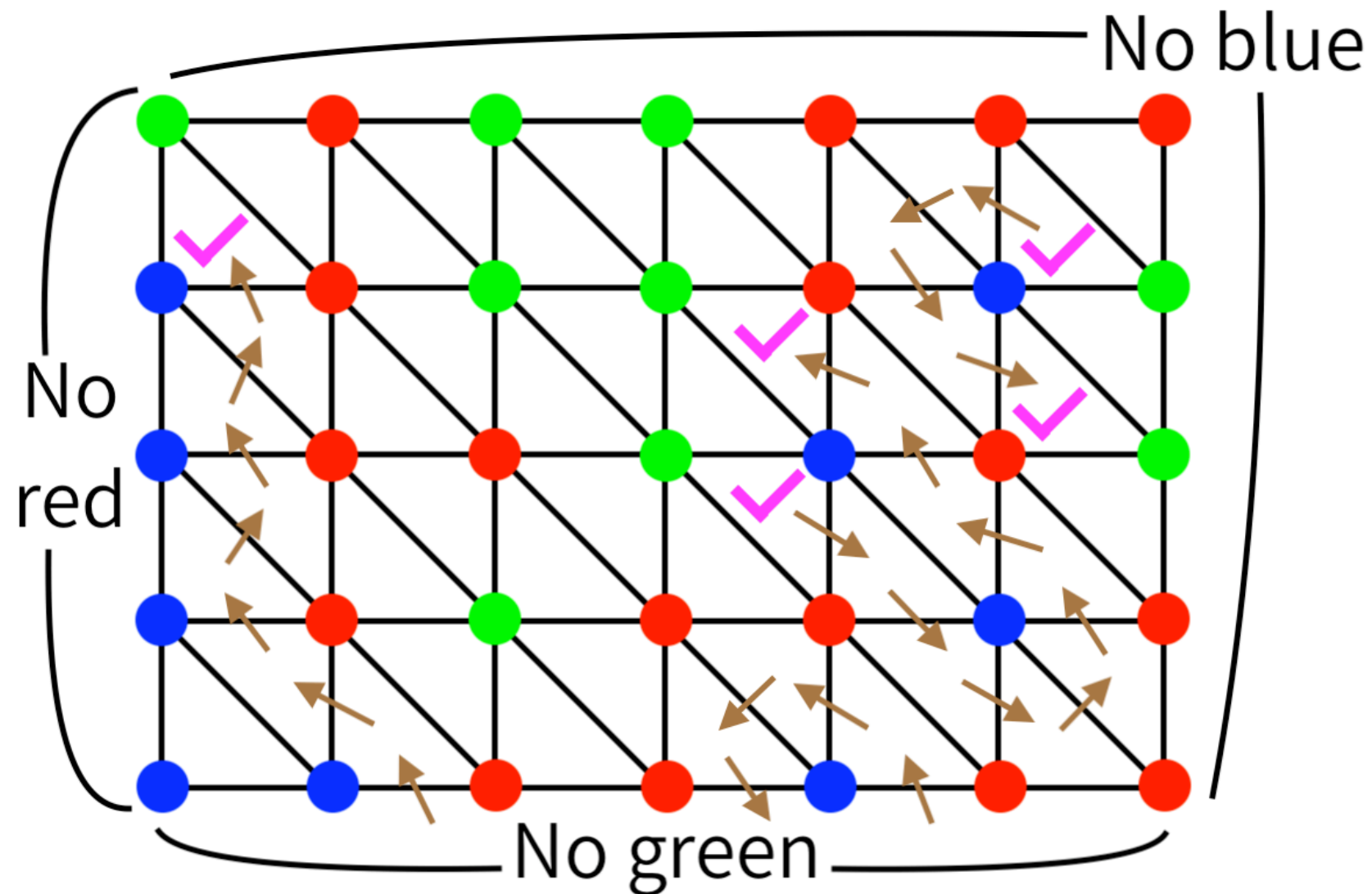
One Special Maze



One Special Maze

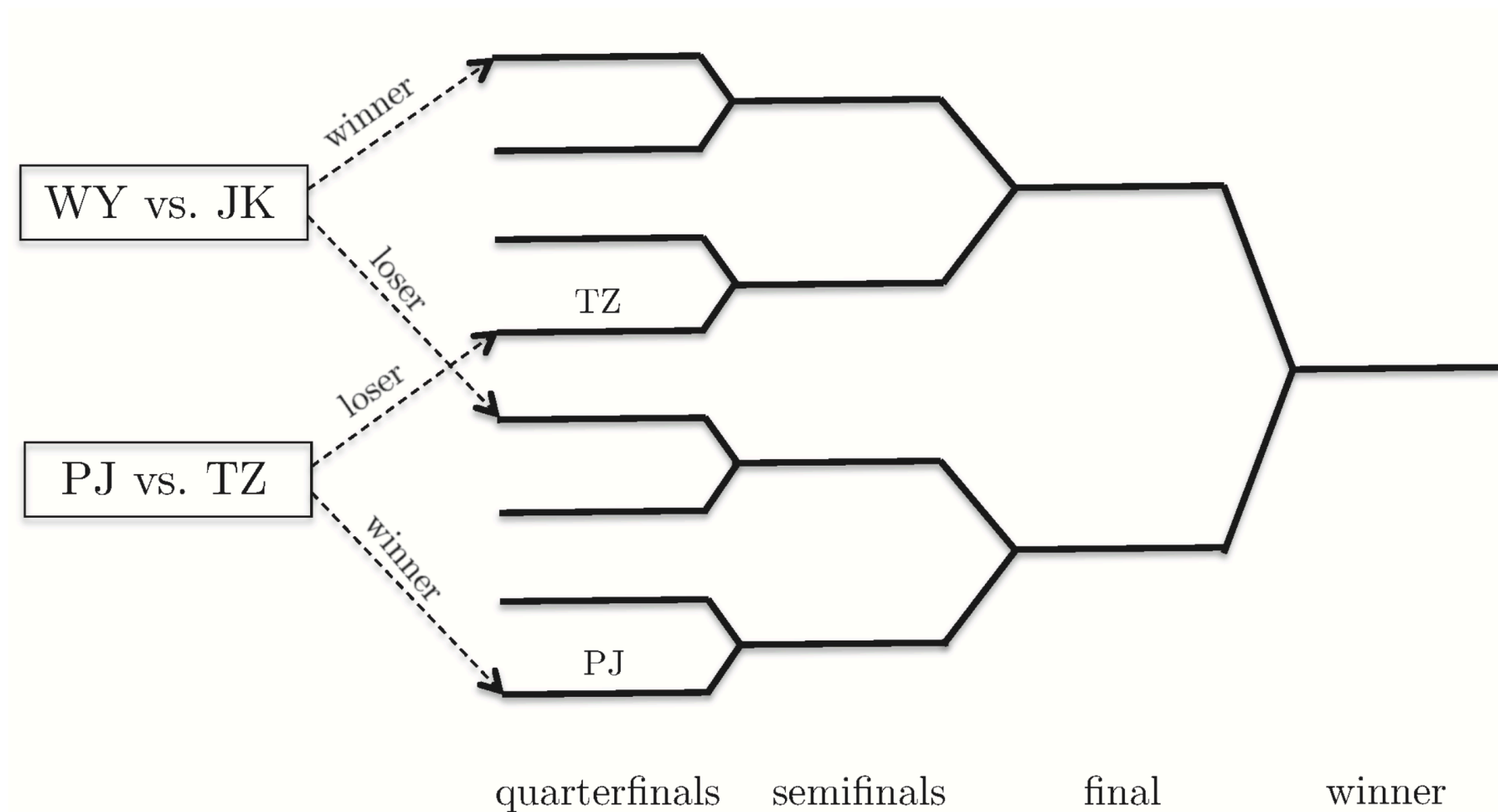


One Special Maze



The Science of Rule-Making

Women's Badminton



Mechanism Design

- In systems with **strategic** participants, **the rules matters**.
- MD: to design rules so that strategic behaviors leads to a **desirable** outcome.
- Examples: Internet Search Auctions, wireless spectrum auctions, the matching of medical residents to hospital, and even kidney exchange.

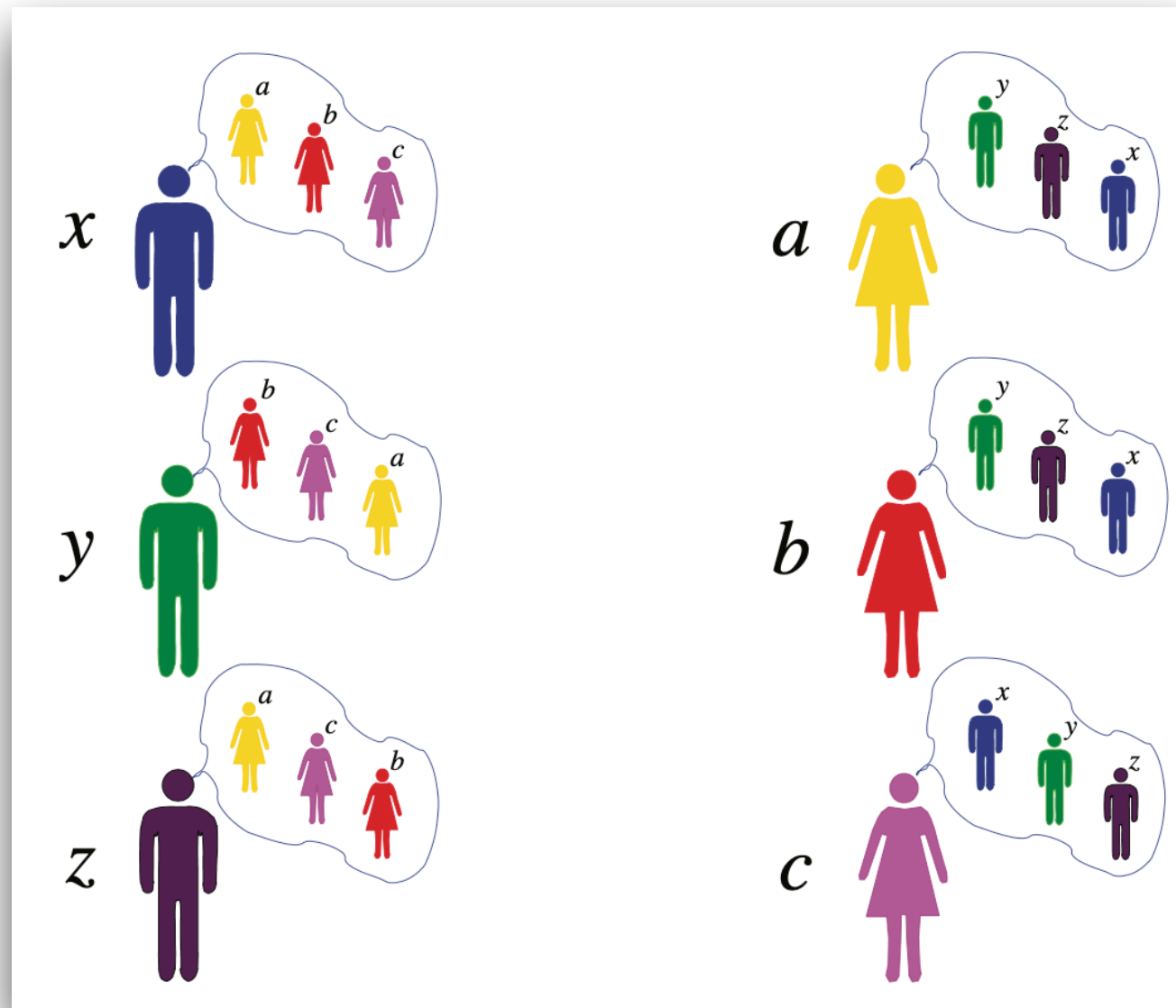
Stable Matching

- n men and m women
- Each one (man/woman) has a private preference list over women/men.
- **Matching**: one-to-one mapping between men and women, denoted by M .
- **Unstable**: two person who are not matched in M , but prefer each other to their partners in M , respectively. Otherwise, we say M is stable.



An unstable pair

An Example



$x \leftrightarrow a, y \leftrightarrow b, z \leftrightarrow c$ is an unstable matching!

Algorithms

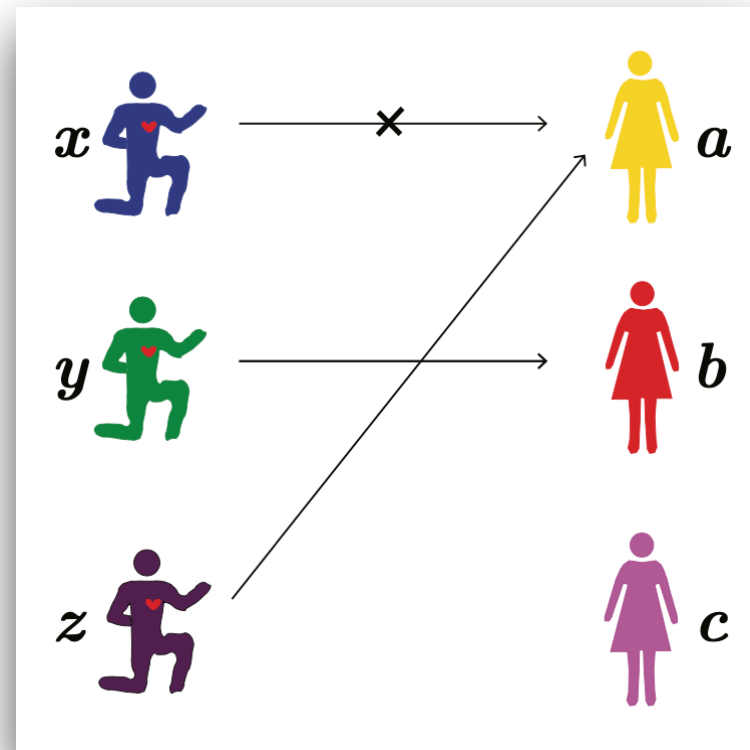
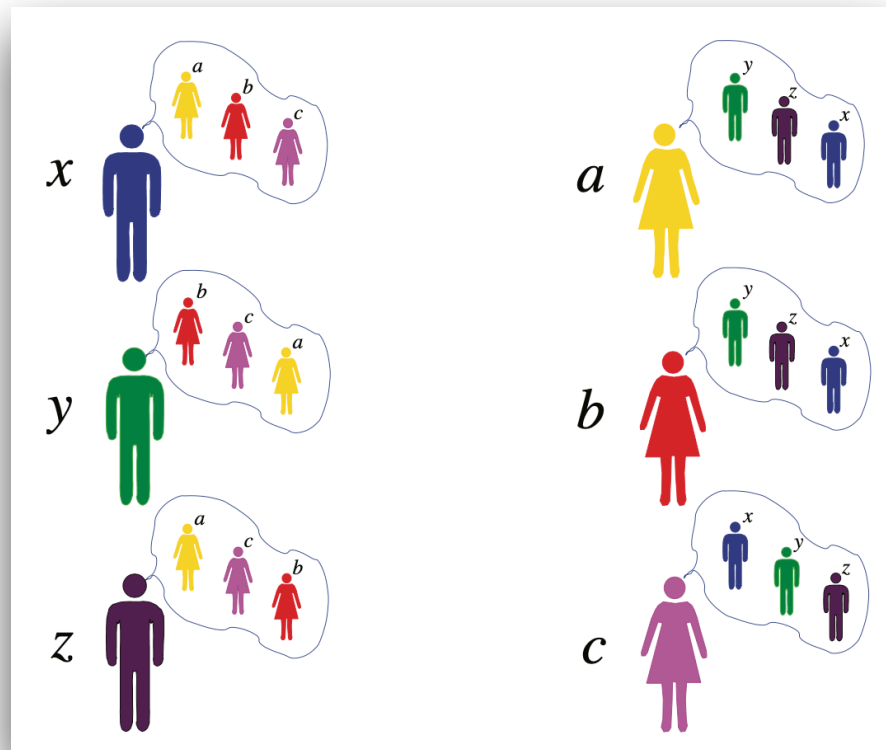
Men-proposing version

1. Initially all men and women are unmatched.
2. Each man proposes to his most preferred woman who has not rejected him yet (or gives up if he's been rejected by all women).
3. Each woman is **tentatively** matched to her favorite among her proposers and rejects the rest.
4. Repeat steps (2) and (3) until a round in which there are no rejections. At that point the tentative matches become final.

- If a man is tentatively matched to a woman in round k and the algorithm doesn't terminate, then he necessarily proposes to her again in round $k + 1$.
- We can also describe the women-proposing algorithm.

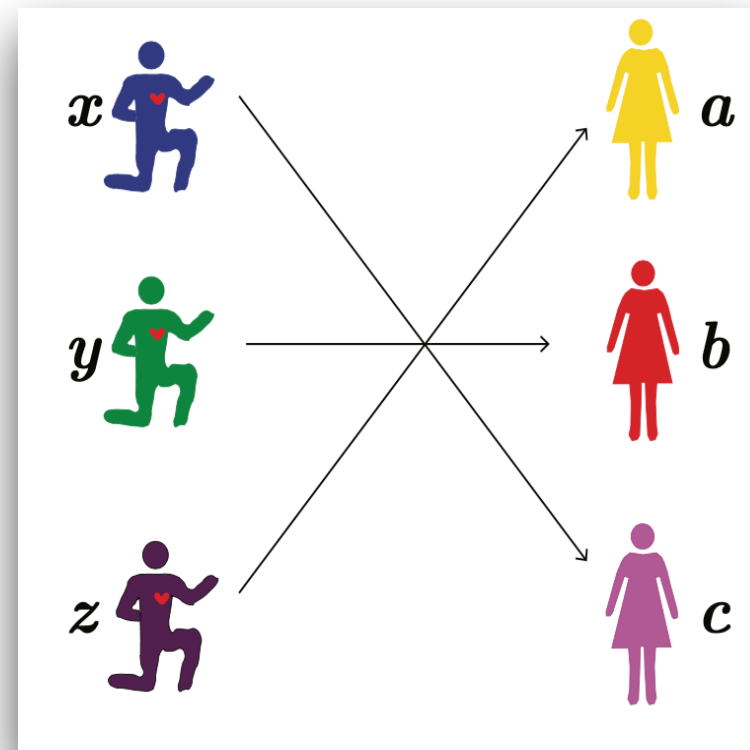
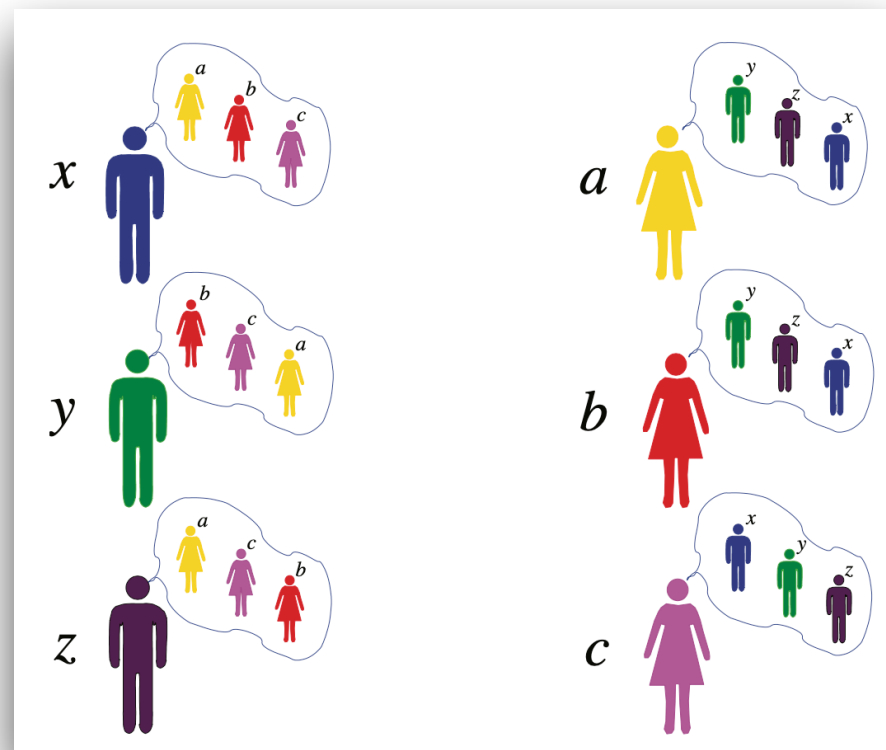
Recall

The last example...



Recall

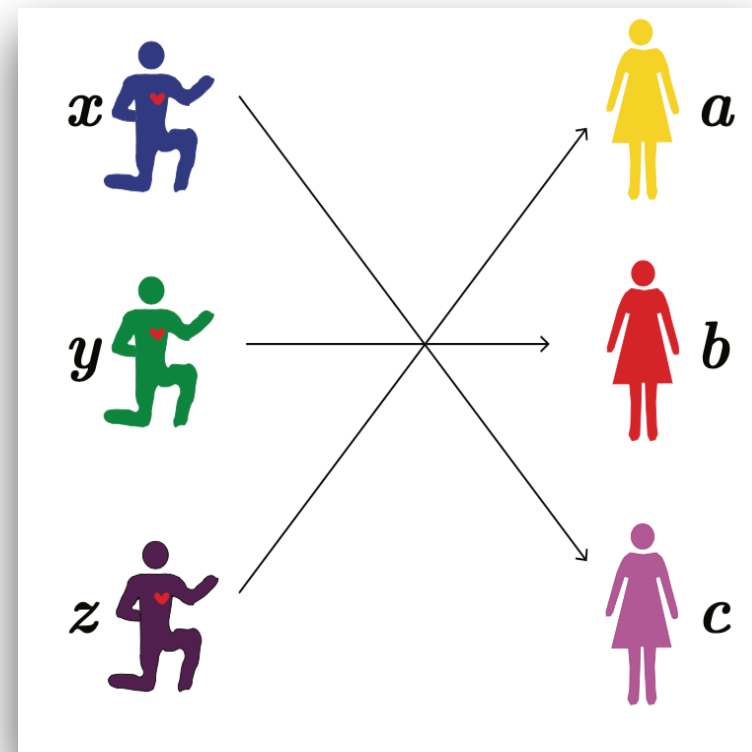
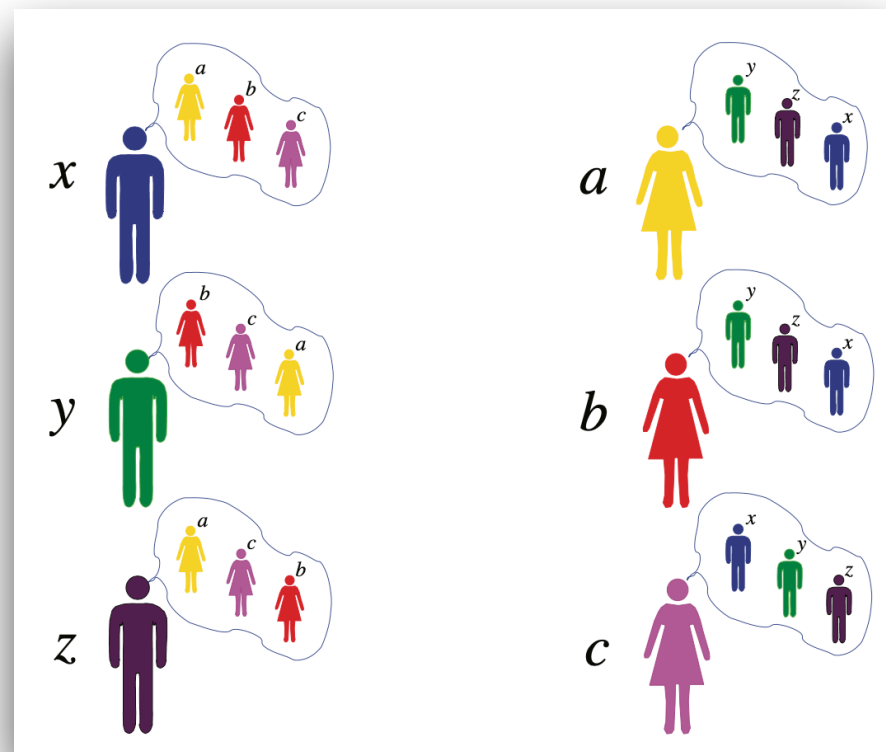
The last example...



The men-proposing algorithm yields a stable matching.

Recall

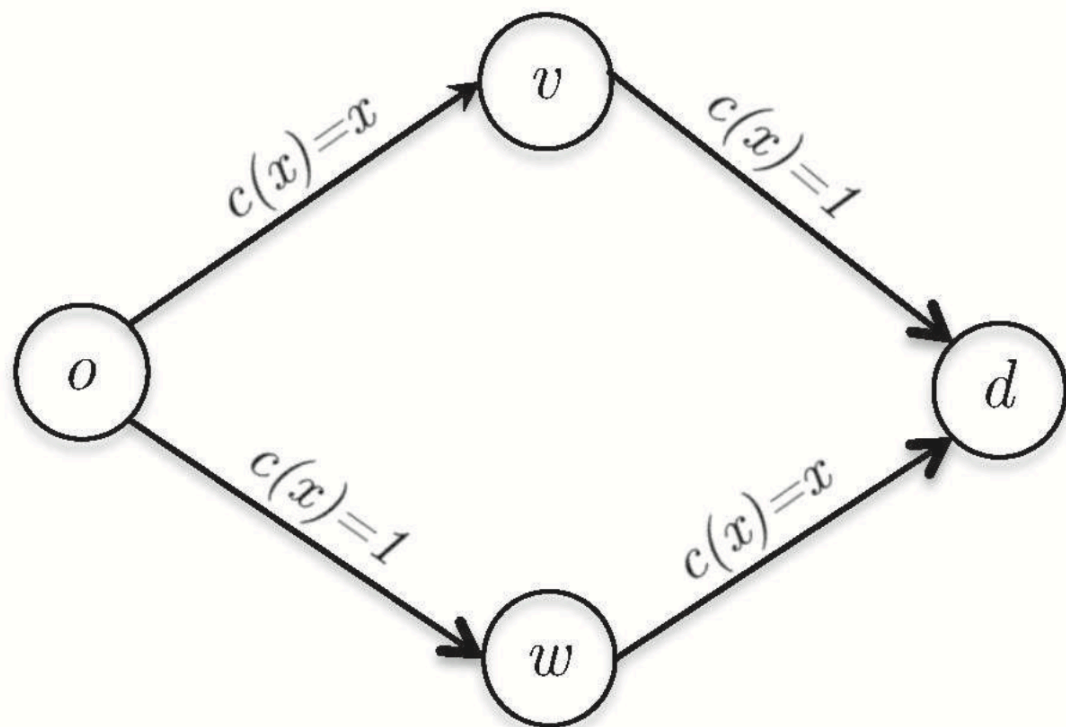
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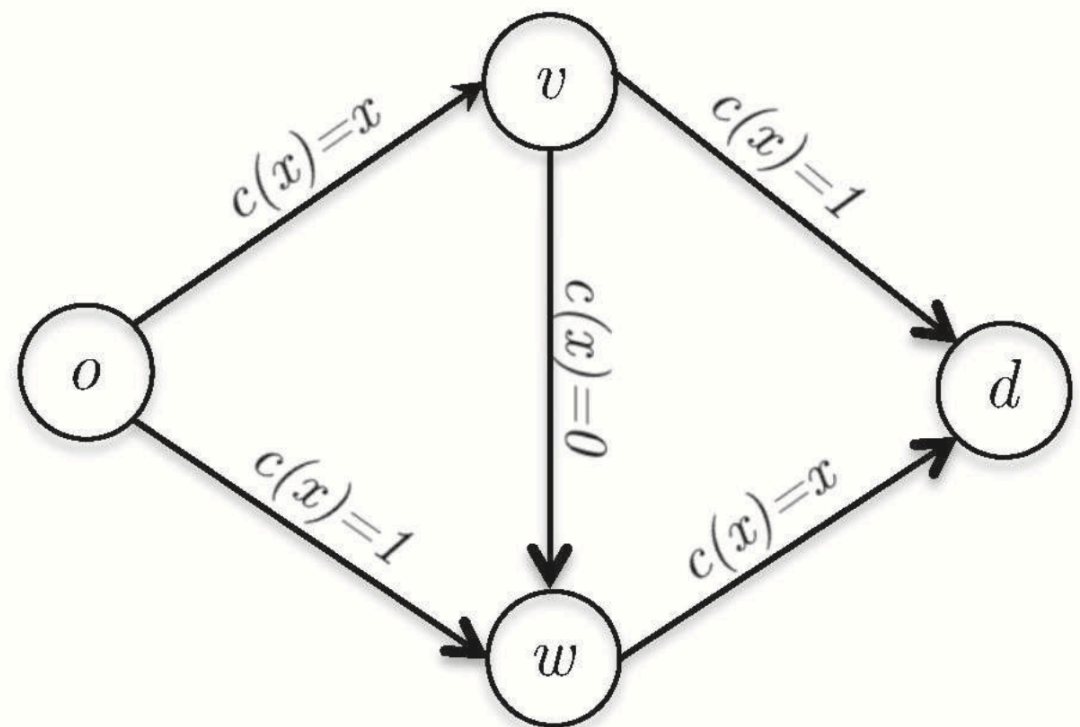
The men-proposing algorithm yields a stable matching.

Each man is matched to his most preferred attainable woman.

Braess's Paradox



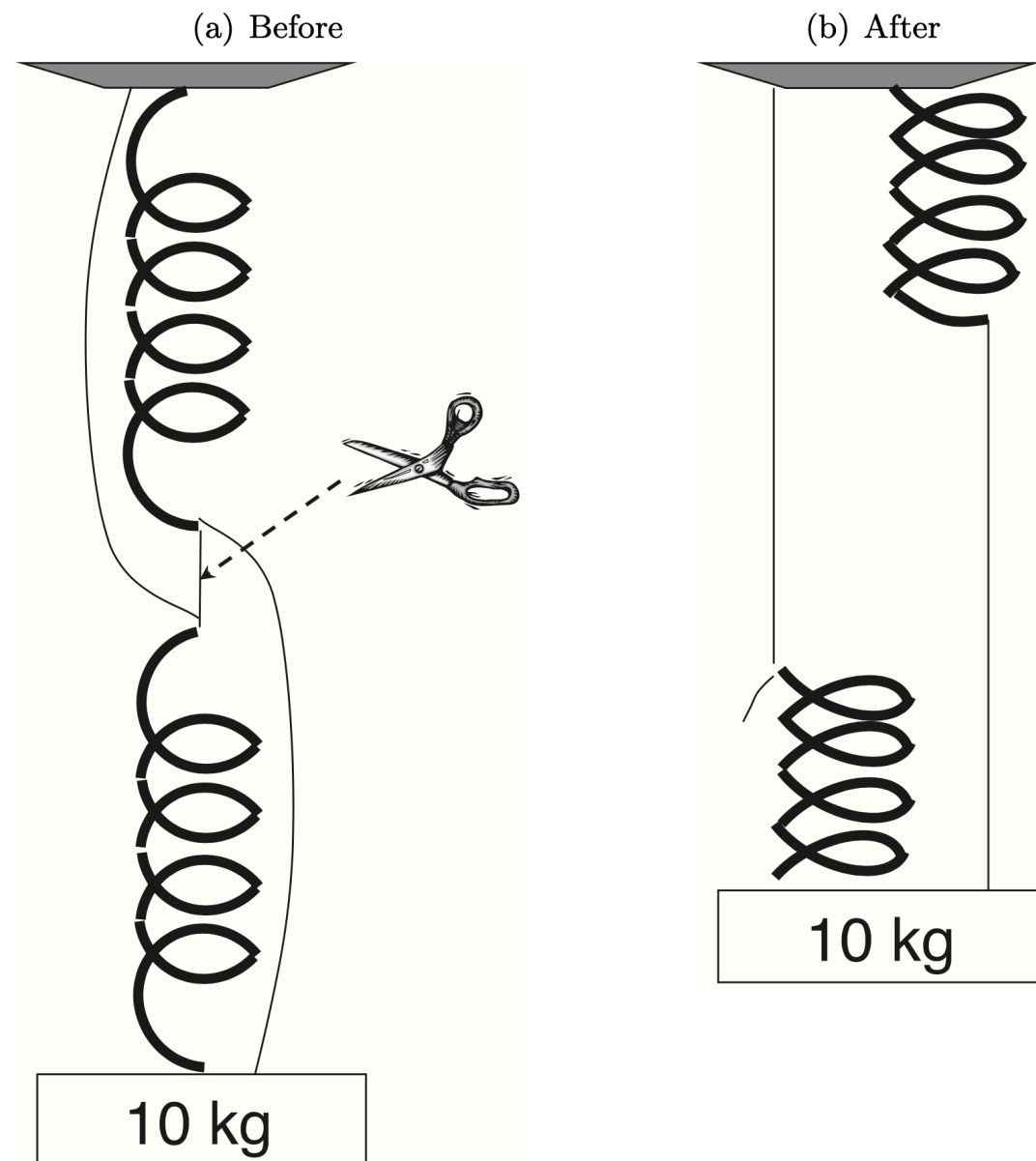
(a) Initial network



(b) Augmented network

We define the **price of anarchy** (POA) as the ratio between the system performance with strategic players and the best-possible system performance.

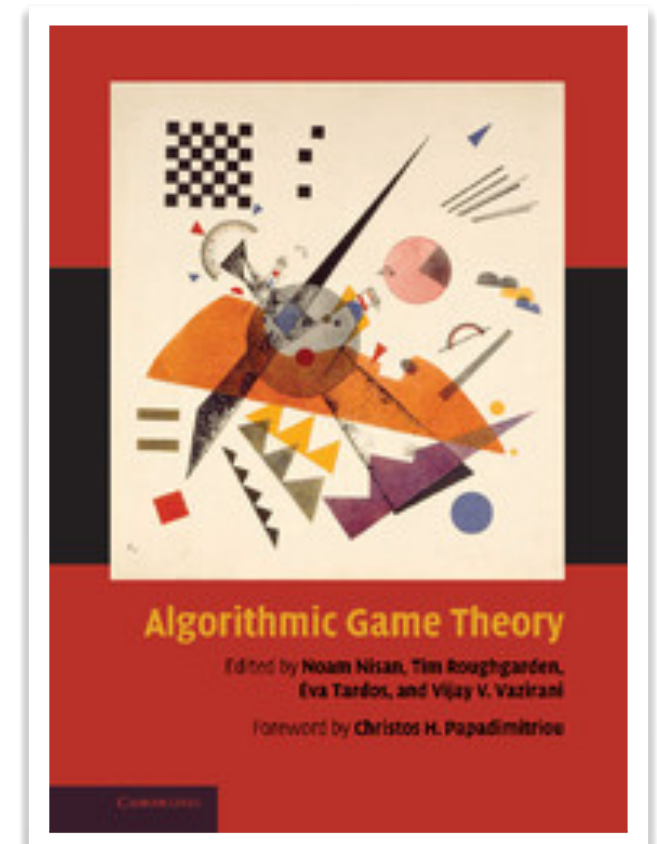
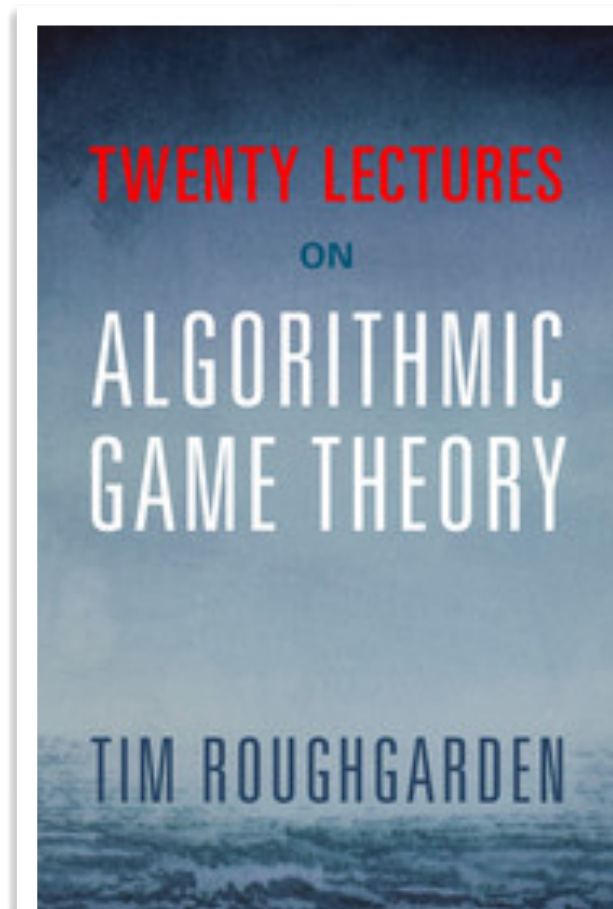
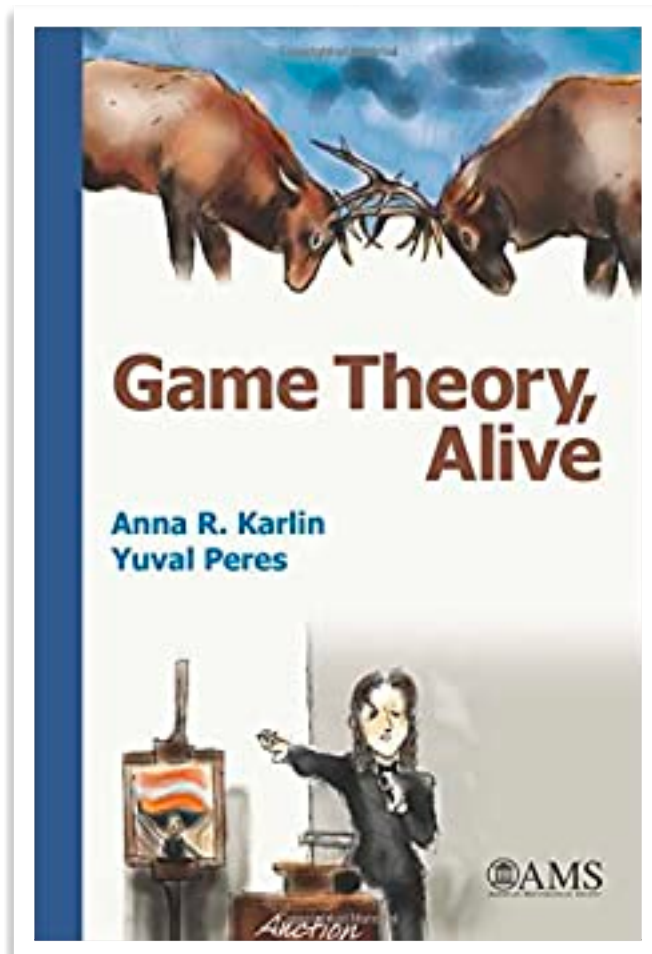
Strings and Springs



Summary

- Lots of GT examples in our life
 - Properties, analysis and prediction
- Need assumptions to study the model
 - personal rationality, or incentive
- Through the computational lens, aka **algorithm + complexity**

Reference



(All these books have online versions.)

1. <https://homes.cs.washington.edu/~karlin/GameTheoryBook.pdf>
2. <http://timroughgarden.org/f13/f13.pdf> (Has Chinese version ...)
3. <https://www.cs.cmu.edu/~sandholm/cs15-892F13/algorithmic-game-theory.pdf>



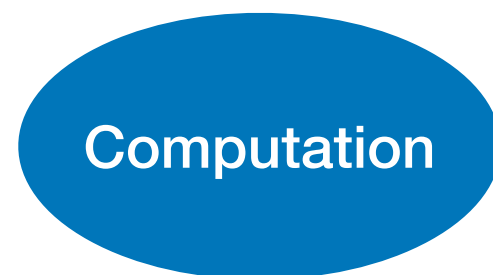
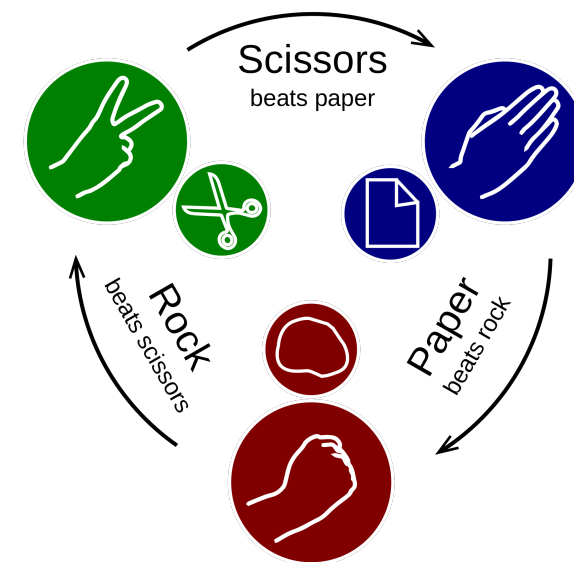
PRISONER'S DILEMMA
 C = Cooperate D = defect (don't Cooperate)

	C	D
C	1, 1	-10, 0
D	0, -10	-5, -5

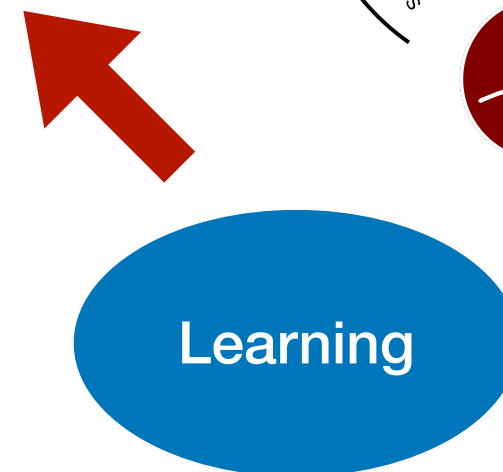
YEARS IN PRISON



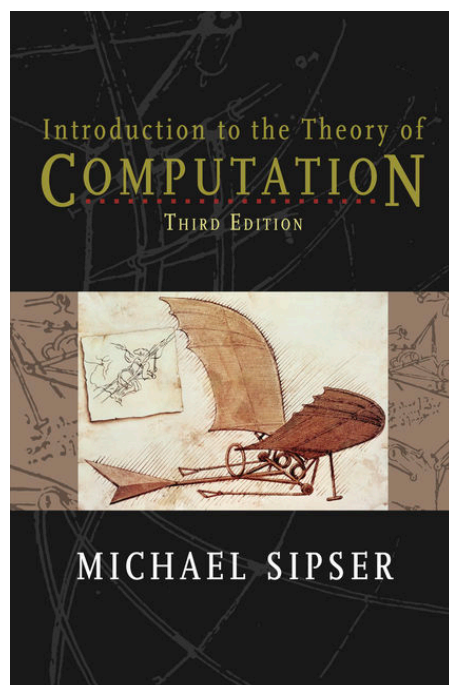
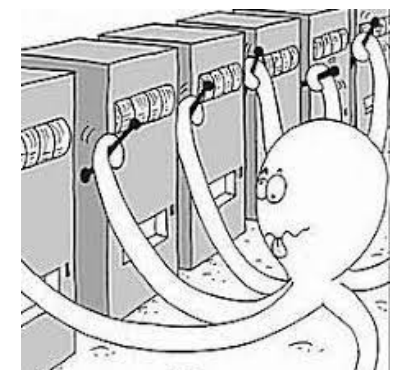
Incentive



Limitations



Randomness



Q&A?

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